ArmyForRE at KnowledgeGraphEval 2026: Typed Entity Markers and Imbalance-Aware Training for Arabic Relation Extraction

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Abstract

In this paper, we present a relation extraction system developed for Subtask 2 of KnowledgeGraphEval 2026. The system fine-tunes a single ARBERTv2 encoder on inputs with typed entity markers that expose entity boundaries and types, also classifies each entity pair from the encoder's classification token (`[CLS]`) state concatenated with the states at the subject and object start markers. It is trained over the full 41-label space, including *no-relation*, with focal loss, class-balanced sampling, label smoothing, and *no-relation* down-weighting to counter severe class imbalance. Our submission ranked third of seven teams with a micro-F1 of 0.9332, 0.23 points above the official baseline, which fine-tunes the same encoder. We also report an analysis of seed-dependent behaviour on fine-grained geopolitical relations from our earlier development experiments and discuss the limitations of our evaluation.

1 Introduction

Knowledge graphs underpin semantic search, question answering, recommendation, and Retrieval-Augmented Generation (RAG) pipelines built on large language models (Hogan et al., 2021; Peng et al., 2024). Their structured (subject, relation, object) representation offers a clear advantage over plain text for downstream reasoning and retrieval.

Relation Extraction (RE), identifying and classifying semantic relations between entities, is therefore a crucial step in constructing knowledge graphs from text. An inaccurate RE stage propagates directly into an incomplete or erroneous knowledge graph, degrading downstream applications such as question answering and GraphRAG (Zhong et al., 2023; Zhao et al., 2024).

Arabic RE research still lags well behind Arabic NER: most prior work has focused on entity recognition, leaving large-scale RE resources scarce (Aljabari et al., 2025b). Arabic's rich morphology, semantic ambiguity, and flexible word order add further difficulty to extracting relations accurately.

Aljabari et al. (2025b) recently introduced WojoodRelations, currently the largest Arabic RE dataset, built on top of the Wojood corpus with fine-grained annotations across 40 semantic relation types, each instance structured as a (subject, relation, object) triple. Its release opens new possibilities for Arabic RE research.

KnowledgeGraphEval 2026 Subtask 2 adopts WojoodRelations directly (Aljabari et al., 2026): gold entity spans are provided, so systems classify the relation between each entity pair into one of 40 types or assign *no-relation* when no connection exists, evaluated by micro-F1.

In this paper, we present our Relation Extraction system for Subtask 2. Our system frames RE as sentence-level classification over typed entity markers and fine-tunes a single ARBERTv2 encoder over all 41 labels with imbalance-aware objectives. Our official submission achieved a micro-F1 of 0.9332, placing third in Subtask 2. Because the official baseline fine-tunes the same encoder, the 0.23-point difference reflects our input representation and training choices taken together. The top of the leaderboard is tightly clustered (Table 2), so we present our results as a competitive, reproducible configuration rather than as evidence that any single component is decisive. Our contributions are:

- Typed entity markers combined with entity-start pooling as an input representation for Arabic RE.
- A training setup over the full label space that keeps *no-relation* predictable while countering class imbalance (focal loss, class-balanced sampling, label smoothing, *no-relation* down-weighting), with a complete hyperparameter specification.
- An analysis, from earlier development runs, of how random seeds change the behaviour

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Baseline	P	R	Micro F_1
ArBERTv2 (fine-tuned)	88.91	88.65	93.09
GPT-4o (few-shot RAG)	89.25	83.66	85.78

Table 1: Official Subtask 2 baselines on the held-out test split (Aljabari et al., 2026): a fine-tuned ArBERTv2 classifier (Abdul-Mageed et al., 2021) and a few-shot GPT-4o (OpenAI, 2023) RAG pipeline. The corpus covers 40 relation types across 17,381 train / 2,074 validation / 4,386 test instances.

of XLM-RoBERTa-large on closely related geopolitical relations (§5.2).

2 Background

The KnowledgeGraphEval 2026 shared task decomposes Arabic knowledge graph construction into cross-domain named entity recognition (Subtask 1: AdaptNER) and relation extraction between identified entities (Subtask 2) (Aljabari et al., 2026).

For Subtask 2, models are trained and evaluated on the WojoodRelations corpus, which extends the Wojood NER corpus (Jarrar et al., 2022) with annotated semantic relations between entities (Aljabari et al., 2025b). Each of the 40 fine-grained relation types is represented as a structured (subject, relation, object) triple. The released data comprises 17,381 training instances, 2,074 validation instances, and a held-out test set of 4,386 instances, with the official evaluation reported as micro-F_1 on the test split.

Task challenges are: (i) *Fine-grained relation classification,* distinguishing among 40 closely related categories, or correctly predicting *no-relation*, requires deep contextual and syntactic understanding of Arabic constructions. (ii) *Interaction with nested entities,* systems must reason over relations between overlapping or nested entities, such as a geopolitical entity nested inside an organization (e.g., "Cairo" inside "Cairo University"). (iii) *Evaluation rigorm* under the official scoring, a single boundary or classification error penalizes both precision and recall simultaneously, which explains the gap between the fine-tuned and few-shot baselines in Table 1.

3 System Overview

3.1 Task Formulation

Given a sentence s and an entity pair $(\mathbf{e}_{\text{subj}}, \mathbf{e}_{\text{obj}})$ with their named entity types, we predict a label $r \in \mathcal{R}$, where $\mathcal{R}$ contains the 40 relation types plus *no-relation* ($|\mathcal{R}| = 41$). The system can therefore decline to assert a relation whenever none of the 40 types holds. The prediction is conditioned on both the textual context and the entity types via typed markers.

3.2 Typed Entity Markers

Following Zhou and Chen (2022), we insert typed entity boundary markers directly into the input text. For a subject entity of type $\texttt{T}_\texttt{s}$ and object entity of type $\texttt{T}_\texttt{o}$, we wrap each entity span with special tokens:

$$\texttt{[SUBJ-T}_\texttt{s}\texttt{]}\ \textit{entity}\ \texttt{[/SUBJ-T}_\texttt{s}\texttt{]}$$

$$\texttt{[OBJ-T}_\texttt{o}\texttt{]}\ \textit{entity}\ \texttt{[/OBJ-T}_\texttt{o}\texttt{]}$$

This produces 60 typed special tokens (subject/object $\times$ 15 types $\times$ open/close) and four generic fallback markers, for 64 markers in total. They are added to the tokenizer vocabulary, and the encoder embedding matrix is resized; the new embeddings are initialized from a normal distribution matching the mean and covariance of the existing embeddings. The markers identify entity boundaries and expose entity types directly in the input sequence.

3.3 Model Architecture

We encode the marked sentence with ARBERTv2 (Abdul-Mageed et al., 2021) (`UBC-NLP/ARBERTv2`, 163M parameters). For the classification representation, we use `cls_entity_start` pooling. We extract the hidden states at the subject-start marker $\mathbf{h}_{\text{ss}}$ and object-start marker $\mathbf{h}_{\text{os}}$, and concatenate them with $\mathbf{h}_{\text{cls}}$, the final hidden state of the classification token `[CLS]` that BERT-style encoders prepend to every input as an aggregate representation of the whole sequence:

$$\mathbf{r} = [\mathbf{h}_{\text{cls}}\ ;\ \mathbf{h}_{\text{ss}}\ ;\ \mathbf{h}_{\text{os}}] \in \mathbb{R}^{3H}$$

This $3H$-dimensional vector is passed through dropout (0.1) and a linear classifier that produces logits over the 41 labels.

3.4 Training

Handling of *no-relation*. The training set contains 17,381 entity pairs, of which 8,092 (46.6%) are labeled *no-relation*. We keep all of them and treat *no-relation* as a 41st class. The task asks systems to predict *no-relation* whenever none of the

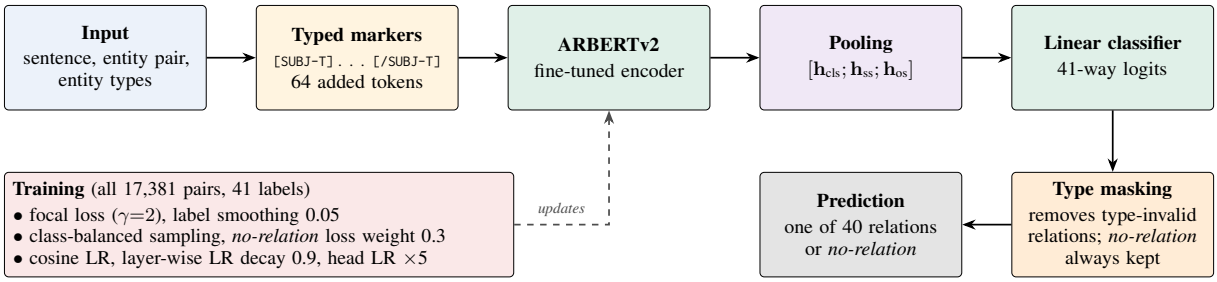


Figure 1: Our RE pipeline. Typed entity markers are inserted around the subject and object, the sentence is encoded by a single fine-tuned ARBERTv2 encoder, and the state of the classification token (`[CLS]`) is concatenated with the states at the subject and object start markers and classified into 41 labels (40 relations plus *no-relation*). Training (bottom left) uses all pairs, including *no-relation*, with imbalance-aware objectives. At inference, ontology-based type masking removes relations whose domain or range is incompatible with the entity types; *no-relation* is never masked.

40 relations holds, and a model that cannot output this label would produce a false positive for every unrelated pair. Training on negative pairs also lets the classifier learn to separate relations from the absence of a relation, not only from one another. Because *no-relation* is by far the largest class, we down-weight its contribution to the loss by a factor of 0.3.

Class-balanced sampling. The 40 relation types are also highly skewed. We draw training batches with a weighted random sampler whose weights are inverse class frequencies, so that each label is sampled with approximately equal probability in every epoch.

Focal loss and label smoothing. We replace standard cross-entropy with focal loss (Lin et al., 2017):

$$\mathcal{L}_{\text{focal}} = -(1 - p_t)^{\gamma} \log p_t$$

with $\gamma = 2.0$, where p_t is the predicted probability of the gold label. The $(1 - p_t)^{\gamma}$ factor down-weights easy, well-classified examples and focuses training on hard instances, which is intended to benefit rare relation types. We additionally apply label smoothing ($\epsilon = 0.05$) to reduce over-confident predictions.

Optimization. We train for 10 epochs with AdamW, a base learning rate of 2×10^{-5}, a batch size of 16 with 2 gradient-accumulation steps (effective batch size 32), and a cosine learning-rate schedule. We apply layer-wise learning-rate decay of 0.9 (each encoder layer receives 0.9 times the learning rate of the layer above it) and give the classification head five times the base learning rate. Training uses mixed precision (fp16). All hyperparameters are listed in Appendix A.

Model selection. We hold out 10% of the 17,381 training pairs as an internal dev set (15,643 train / 1,738 dev, random seed 42) and keep the checkpoint with the highest internal-dev micro-F1 over all 41 labels. Because every pair receives exactly one label, this score equals accuracy, and it is not necessarily computed the same way as the official score.

3.5 Inference-Time Type Constraints

The domain (allowed subject type) and range (allowed object type) of each relation are taken from WojoodOntology (Aljabari et al., 2025a), which formalizes the 40 WojoodRelations relation types (Aljabari et al., 2025b) as OWL object properties with domain–range constraints. For example, *capital_of* requires a GPE subject and a GPE object. At inference time we apply hard constraint masking: the logit of every relation whose domain or range is violated by the given entity types is set to $-\infty$ before the softmax. *No-relation* has no domain or range restriction and is never masked. If either entity type is unknown, no constraints are applied.

4 Experimental Setup

Data. We train on all 17,381 annotated entity pairs of WojoodRelations (Aljabari et al., 2025b), including the 8,092 *no-relation* pairs, across 41 labels. The official test set contains 4,386 pairs spanning 2,673 sentences.

Hardware. The submitted model was trained on a single NVIDIA T4 GPU (16 GB) on Google Colab; one epoch takes about four minutes. Earlier development runs (§5.2) used an NVIDIA RTX 3070 Ti (8 GB).

Implementation. Our system is implemented in

Rank	Team	Micro-F1
1	RabEt	0.9406
2	NAMAA	0.9348
3	**ArmyForRE (ours)**	**0.9332**
4	Wasl	0.9328
–	Baseline (ArBERTv2)	0.9309
5	McMasterNLP	0.9295
6	ISNAD	0.9255
7	YafaRE	0.9192

Table 2: Final official results for KnowledgeGraphEval 2026 Subtask 2 (held-out test set), as announced by the organizers.

System	Evaluation	Score
ArBERTv2 baseline	Official test	0.9309
Ours (submitted)	**Official test**	**0.9332**
Ours (submitted)	Internal dev	0.9462
Variant: no *nr* weight, 6 ep.	Internal dev	0.9419

Table 3: Official test micro-F1 and internal-dev scores (41-label micro-F1, equal to accuracy; 1,738 pairs). Internal-dev and official scores come from different splits and are not directly comparable. *nr*: *no-relation*.

PyTorch using HuggingFace Transformers.

5 Results and Analysis

5.1 Main Results

Table 2 shows the final official Subtask 2 ranking. Our submission ranked third with a micro-F1 of 0.9332, 0.23 points above the ArBERTv2 baseline, 0.74 points behind the first-ranked system, and 0.04 points ahead of the fourth-ranked system. Given these small margins and the absence of significance testing, the ranking among the top systems should be interpreted with caution.

Table 3 places the official score next to the internal-dev scores. The submitted model reached its best internal-dev score of 0.9462 at epoch 6 of 10; the score rose from 0.8806 after the first epoch and stayed between 0.9452 and 0.9462 for the remaining epochs. We also trained one unsubmitted variant without *no-relation* down-weighting and with a shorter, 6-epoch schedule, which reached 0.9419. Since this variant changes two settings at once and each configuration was run with a single seed, the 0.43-point difference is not evidence for either setting in isolation.

The internal-dev score is 1.3 points above the official score. The internal dev set is a random sample of the training pool that shares its distribution and was used for checkpoint selection, so it is an optimistic estimate.

5.2 Earlier Development Runs

Before settling on the submitted model, we built a 40-label system that removed *no-relation* from training and masked it at inference. It combined four XLM-RoBERTa-large (Conneau et al., 2020) runs (seeds 42–45, trained with Adafactor (Shazeer and Stern, 2018), a cosine schedule, and gradient checkpointing), two ARBERTv2 runs, and one AraBERTv0.2-large (Antoun et al., 2020) run in a soft-voting ensemble. We replaced it with the single 41-label model because a system that can never output *no-relation* does not solve the task as defined. The 40-label system is not the submitted model; we report two observations from it because they concern model behaviour rather than the final score.

Seed variance. On a 928-pair internal dev set of positive pairs only, XLM-R-large reached 0.9440 (seed 42) and 0.9655, 0.9666, and 0.9666 (seeds 43–45), a spread of over 2 points under an otherwise fixed configuration. These scores measure discrimination among the 40 relations only and are not comparable to the scores in Table 3.

Capital-of collapse. Seeds 43–45 predicted only 0–15 *capital_of* instances on the test set, compared to 24–25 for seed 42 and for the BERT models. On the internal dev set, these seeds absorbed *capital_of* into the closely related *geopolitical_division* class, a seed-specific collapse between neighbouring classes that share identical entity types (GPE→GPE). In the ensemble, the BERT models partially offset this tendency, bringing the *capital_of* count to 15–17. Since test labels are not available to us, we cannot verify which count is closer to the gold distribution; the analysis shows that models disagree systematically on this class, not which one is correct. We did not repeat this analysis for the submitted single model.

5.3 Error Analysis

Inspection of confusion matrices from our development runs points to two recurring error patterns. We describe them qualitatively, because the underlying counts were not preserved with the submitted model.

Geopolitical confusion. *capital_of* and *geopolitical_division* are confused in both directions. The two relations share identical entity types

(GPE→GPE), so type information cannot separate them, and they often appear in similar syntactic contexts ("X is the capital of Y" vs. "X is a region of Y"). This is the same pair affected by the seed-specific collapse in §5.2.

Affiliation confusion. *employee_of*, *member_of*, and *leader_of* are frequently confused with each other, particularly when the subject is a person and the organization context is ambiguous (e.g., a politician who is both a member of a party and an employee of a government body).

5.4 Missing Ablations

We did not run controlled ablations of the individual components (typed versus untyped markers, focal loss versus cross-entropy, with versus without class-balanced sampling, *no-relation* down-weighting, or type masking), and each configuration was trained with a single seed. Consequently, we cannot attribute the 0.23-point gain over the baseline to any single component, and the component-level claims in this paper are design motivations rather than measured effects. We likewise report micro-F1 only; since focal loss and class-balanced sampling target rare relations, macro-F1 and per-relation scores would be the appropriate evidence for their effect.

6 Limitations

Our system has several limitations. First, the submitted system is a single model trained with a single seed, and our development runs showed seed spreads of over 2 points for XLM-R-large; the official score may therefore partly reflect a favourable seed, and the 0.04-point margin over the fourth-ranked team is well within such variation. Second, we provide no controlled ablations, macro-F1, per-relation results, or significance tests. Third, *no-relation* down-weighting and class-balanced sampling both reduce the influence of the largest class, which may bias the model against *no-relation* where unrelated pairs dominate, such as open extraction over all entity pairs in a document; we did not tune this trade-off or calibrate a *no-relation* threshold. Fourth, our checkpoint-selection metric (41-label accuracy on internal dev) is not necessarily the official metric. Fifth, the seed and *capital_of* analyses come from earlier 40-label models and may not transfer to the submitted model. Finally, type-constraint masking depends on the entity types supplied with each pair; incorrect types would exclude the correct relation.

7 Conclusion

We presented ArmyForRE, an Arabic RE system that fine-tunes a single ARBERTv2 encoder with typed entity markers, entity-start pooling, and imbalance-aware training over the full 41-label space, including *no-relation*. The system achieved 0.9332 micro-F1 on the official test set, placing third in KnowledgeGraphEval 2026 Subtask 2 and 0.23 points above a baseline that fine-tunes the same encoder. Earlier development runs showed that XLM-RoBERTa-large behaves differently across random seeds, notably on *capital_of* versus *geopolitical_division*. Future work includes controlled ablations, multi-seed evaluation, per-relation and macro-F1 analysis, and calibrating the *no-relation* decision for settings with many unrelated pairs.

References

Muhammad Abdul-Mageed, AbdelRahim Elmadany, and El Moatez Billah Nagoudi. 2021. ARBERT & MARBERT: Deep bidirectional transformers for Arabic. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7088–7105, Online. Association for Computational Linguistics.

Alaa Aljabari, Nagham Hamad, Abdellah El Mekki, Muhammad Abdul-Mageed, Imed Zitouni, Sanjay Chawla, and Mustafa Jarrar. 2026. KnowledgeGraphEval 2026: The First Shared Task on Arabic Knowledge Graphs. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Alaa Aljabari, Nagham Hamad, Mohammed Khalilia, and Mustafa Jarrar. 2025a. WojoodOntology: Ontology-driven LLM prompting for unified information extraction tasks. In *Proceedings of The Third Arabic Natural Language Processing Conference*, pages 179–193, Suzhou, China. Association for Computational Linguistics.

Alaa Aljabari, Mohammed Khalilia, and Mustafa Jarrar. 2025b. Wojood^{Relations}: Arabic relation extraction corpus and modeling. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 34342–34360, Suzhou, China. Association for Computational Linguistics.

Wissam Antoun, Fady Baly, and Hazem Hajj. 2020. AraBERT: Transformer-based model for Arabic language understanding. In *Proceedings of the 4th Work-*

shop on Open-Source Arabic Corpora and Processing Tools, with a Shared Task on Offensive Language Detection, pages 9–15, Marseille, France. European Language Resource Association.

Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451, Online. Association for Computational Linguistics.

Aidan Hogan, Eva Blomqvist, Michael Cochez, Claudia d'Amato, Gerard de Melo, Claudio Gutierrez, Sabrina Kirrane, José Emilio Labra Gayo, Roberto Navigli, Sebastian Neumaier, Axel-Cyrille Ngonga Ngomo, Axel Polleres, Sabbir M. Rashid, Anisa Rula, Lukas Schmelzeisen, Juan Sequeda, Steffen Staab, and Antoine Zimmermann. 2021. Knowledge graphs. *ACM Computing Surveys*, 54(4):71:1–71:37.

Mustafa Jarrar, Mohammed Khalilia, and Sana Ghanem. 2022. Wojood: Nested Arabic named entity corpus and recognition using BERT. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 3626–3636, Marseille, France. European Language Resources Association.

Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. 2017. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2980–2988.

OpenAI. 2023. GPT-4 technical report. *arXiv preprint arXiv:2303.08774*.

Boci Peng, Yun Zhu, Yongchao Liu, Xiaohe Bo, Haizhou Shi, Chuntao Hong, Yan Zhang, and Siliang Tang. 2024. Graph retrieval-augmented generation: A survey. *arXiv preprint arXiv:2408.08921*.

Noam Shazeer and Mitchell Stern. 2018. Adafactor: Adaptive learning rates with sublinear memory cost. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 4596–4604. PMLR.

Siyun Zhao, Yuqing Yang, Zilong Wang, Zhiyuan He, Luna Qiu, and Lili Qiu. 2024. Retrieval augmented generation (RAG) and beyond: A comprehensive survey on how to make your LLMs use external data more wisely. *arXiv preprint arXiv:2409.14924*.

Lingfeng Zhong, Jia Wu, Qian Li, Hao Peng, and Xindong Wu. 2023. A comprehensive survey on automatic knowledge graph construction. *arXiv preprint arXiv:2302.05019*.

Wenxuan Zhou and Muhao Chen. 2022. An improved baseline for sentence-level relation extraction. In *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, pages 161–168, Online only. Association for Computational Linguistics.

A Hyperparameters

Setting	Value
Encoder	`UBC-NLP/ARBERTv2`
Pooling	`cls_entity_start`
Labels	41 (incl. *no-relation*)
Optimizer	AdamW
Base learning rate	2×10^{-5}
Head LR multiplier	5.0
Layer-wise LR decay	0.9
LR schedule	cosine
Epochs	10 (best: epoch 6)
Batch × grad. accum.	16×2
Loss	focal, $\gamma = 2.0$
Label smoothing	0.05
no-relation weight	0.3
Sampling	class-balanced
Dropout	0.1
Precision	fp16
Split seed	42
Train / int. dev	15,643 / 1,738
Hardware	NVIDIA T4 (16 GB)

Table 4: Training configuration of the submitted model.